Lab 16: Configure AWS Backup for EC2 & RDS

Lab Objective:

In this lab, you will configure AWS Backup to:

- Protect EC2 instances
- Protect RDS databases
- Create a Backup Plan
- Assign resources using tags
- Perform an on-demand backup

Prerequisites

- AWS account with AWS Backup permissions
- At least:
    - 1 EC2 instance (running)
    - 1 RDS instance (available)

Lab Steps

Step 1: Log in & Select Region

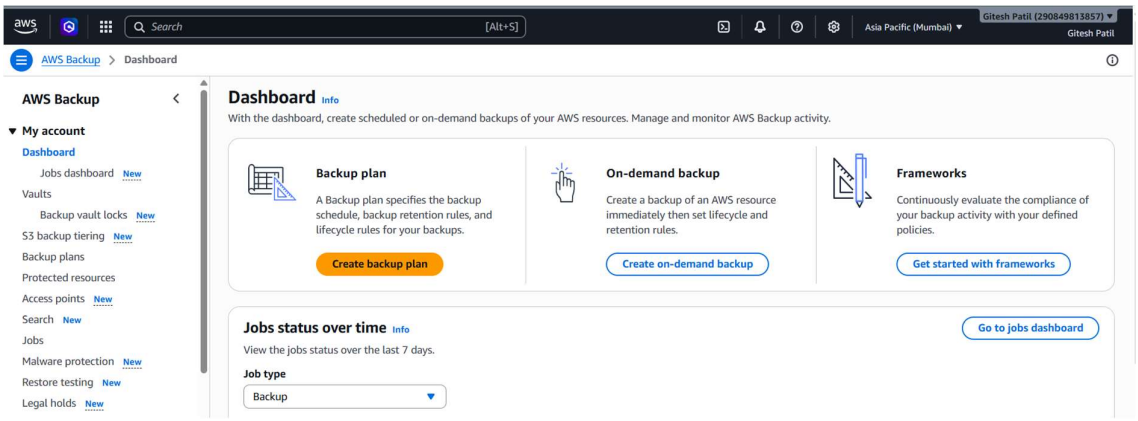
1. Log in to AWS Console

2. Select region (same region where EC2 & RDS exist)

Step 2: Open AWS Backup

1. Search AWS Backup in console

2. Click the service



Step 3: Create a Backup Vault

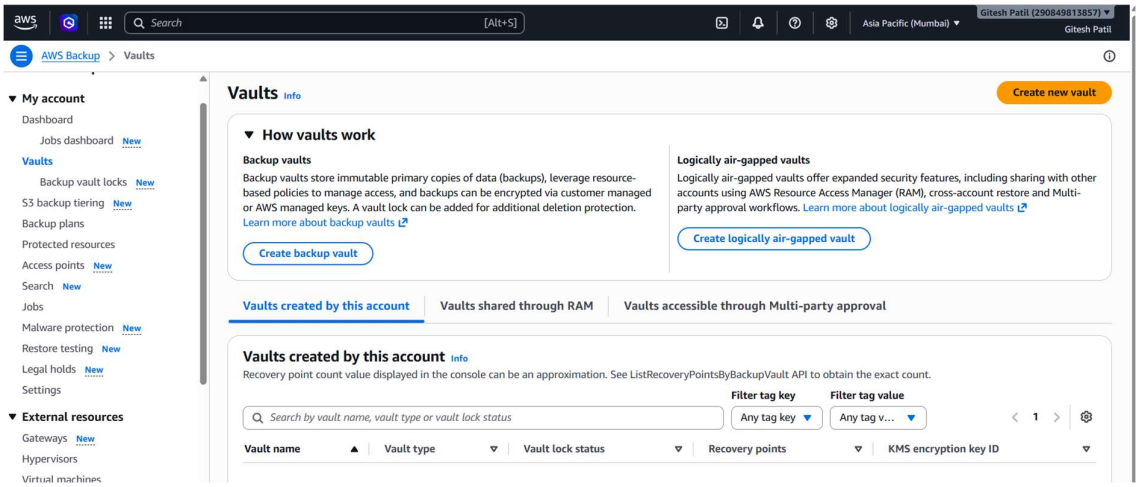
1. In left menu Backup vaults

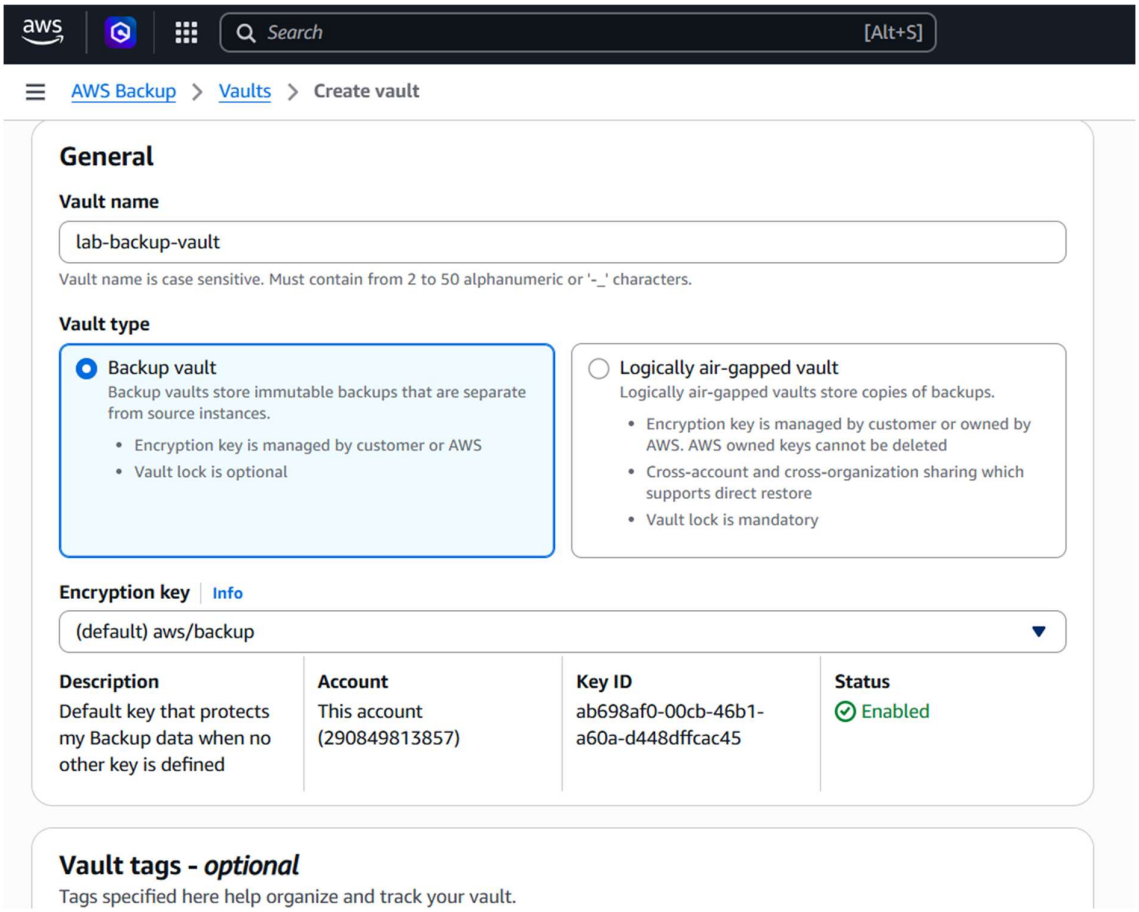
2. Click Create backup vault

3.Enter: Name: lab-backup-vault

4.Keep encryption as default

5.Click Create





Step 4: Tag Resources (EC2 & RDS) For EC2

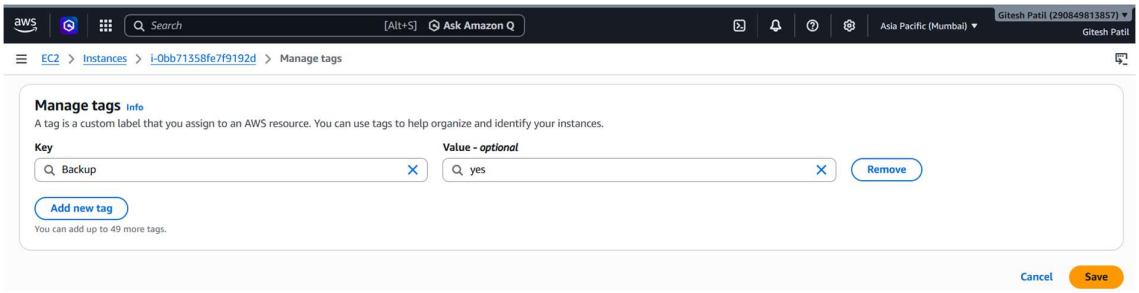
1. Go to EC2 -> Instances

2. Select your instance

3. Click Tags -> Manage tags

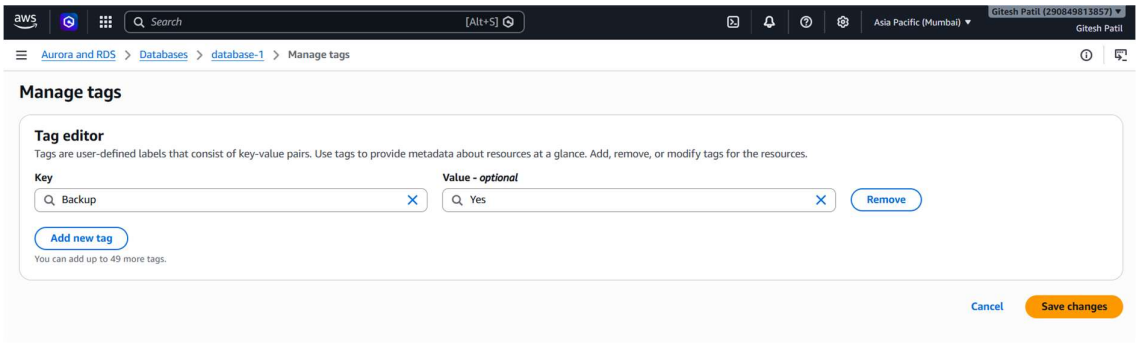
4.Add:

- Key: Backup
- Value: Yes



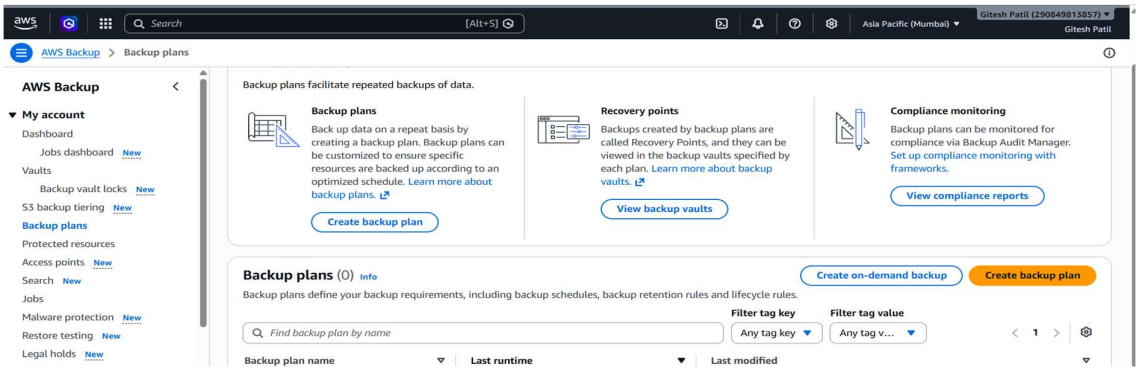
For RDS

1. Go to RDS -> Databases
2. Select DB instance
3. Click Tags
4. Add:
    - Key: Backup
    - Value: Yes



Step 5: Create Backup Plan

1. Go back to AWS Backup
2. Click Backup plans
3. Click Create backup plan



Step 6: Configure Backup Plan

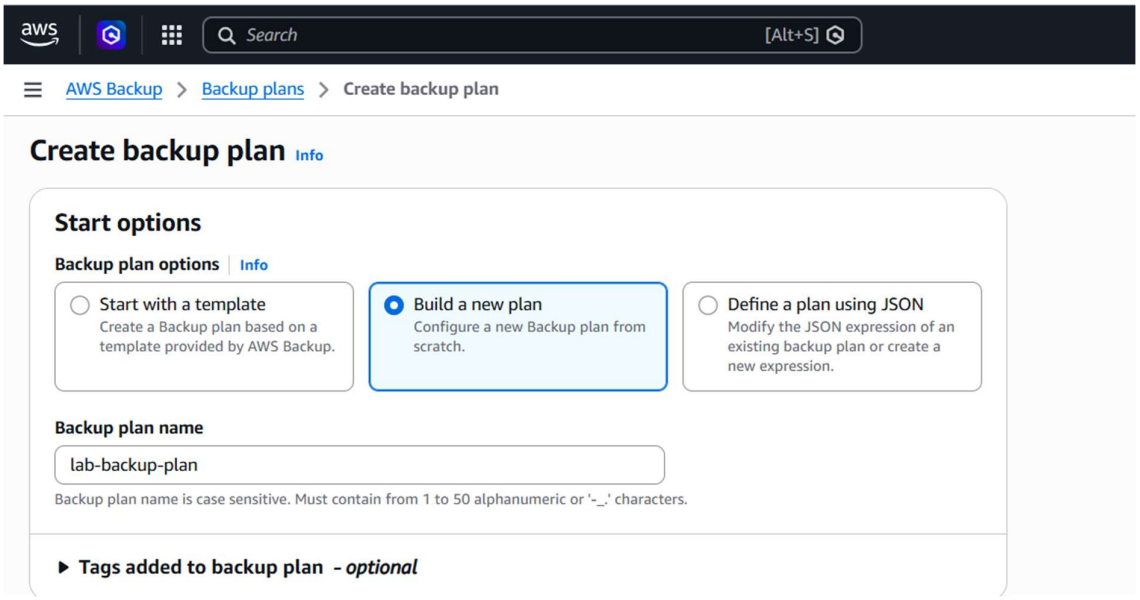
1. Choose: Build a new plan

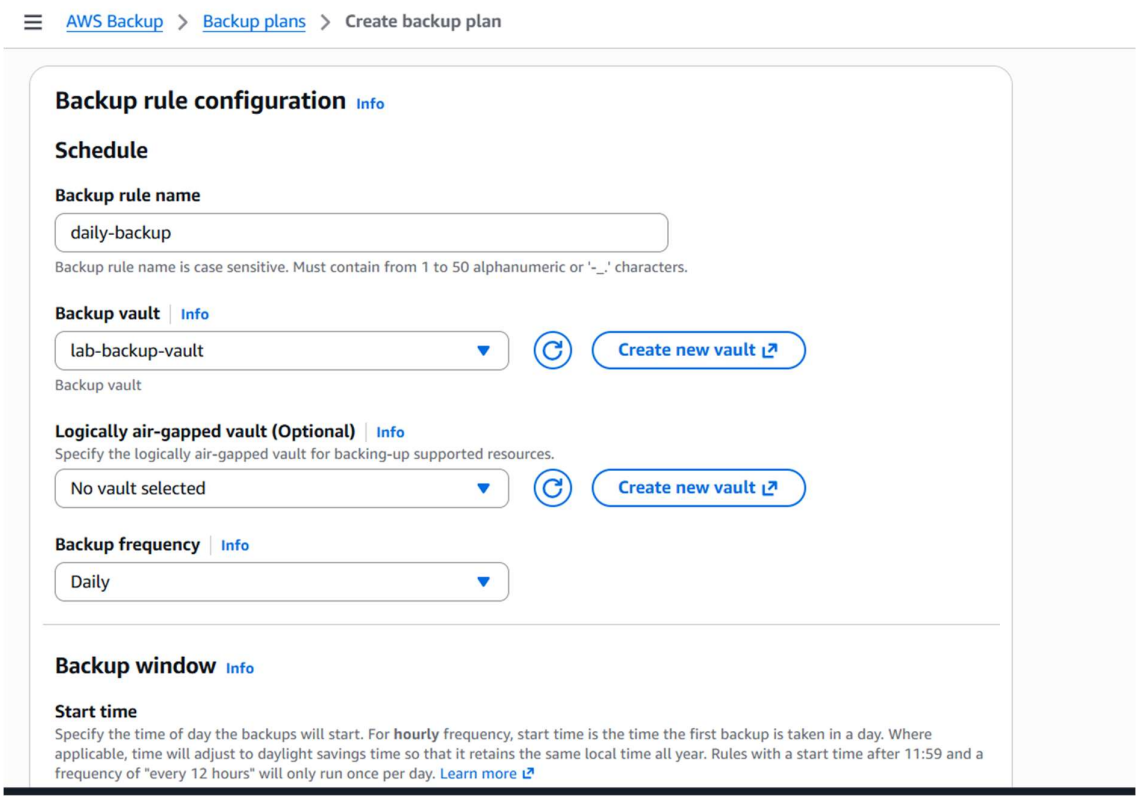
2. Plan name: lab-backup-plan

3.Rule configuration:

- Rule name: daily-backup
- Backup vault: lab-backup-vault
- Frequency: Daily
- Start time: leave default
- Lifecycle: keep default

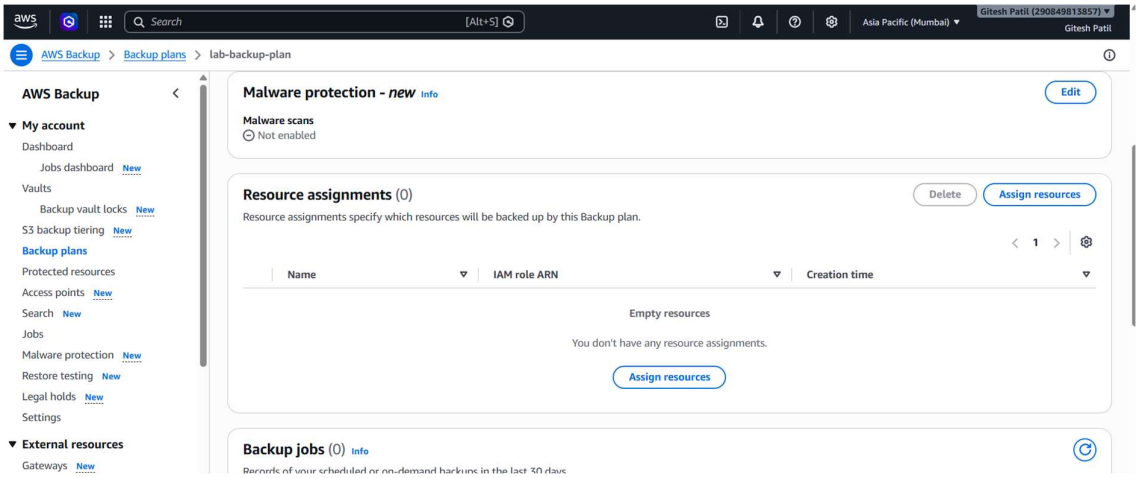
4.Click Create Plan





Step 7: Assign Resources to Backup Plan

1. After plan creation Click Assign resources



2. Assignment name: lab-resource-assignment

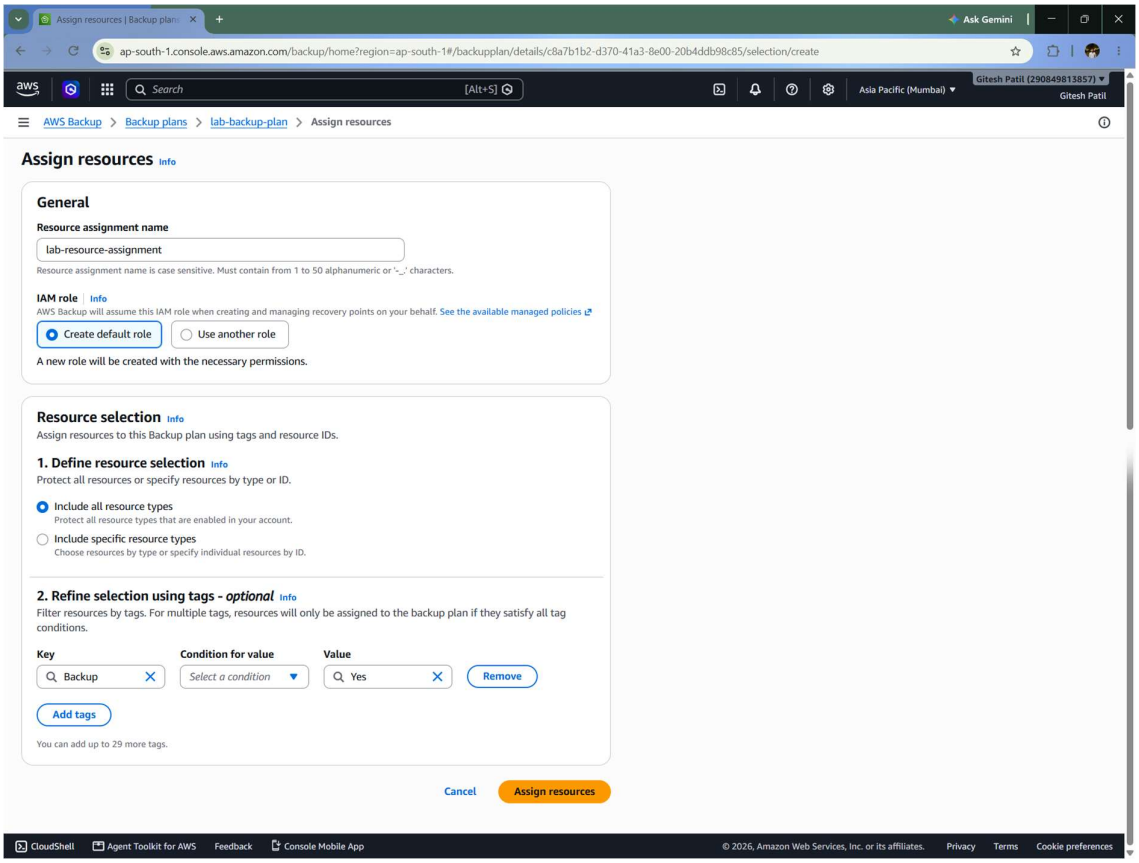
3. IAM role: Choose Default role

4. Resource selection: Choose Refine selection using tags

5. Tag condition:

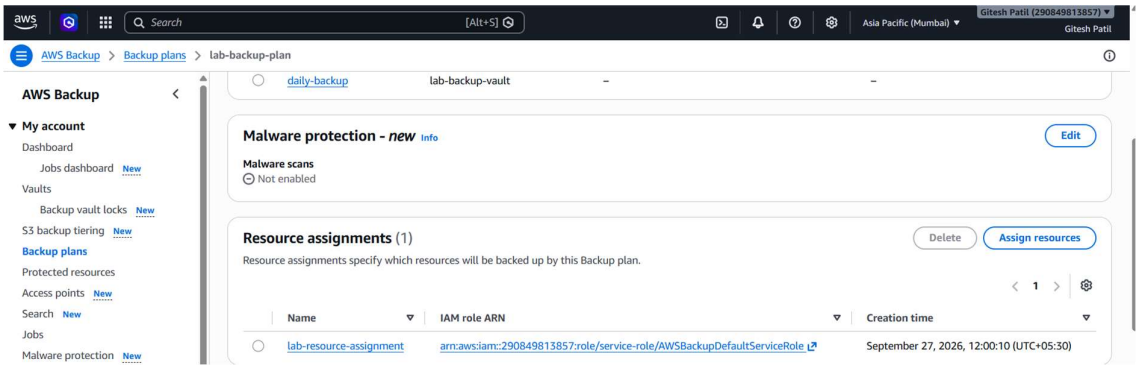
- Key: Backup
- Value: Yes

6.Click Assign resources



Step 8: Verify Resource Assignment

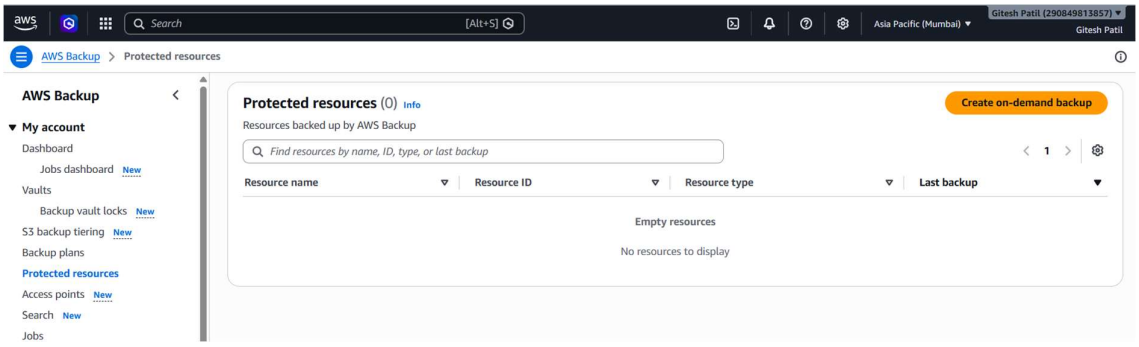
1. Open backup plan
2. Check:
    - Resource assignments tab
    - EC2 & RDS resources listed



Step 9: Run On-Demand Backup (Important)

1. Go to Protected resources
2. Select EC2 instance

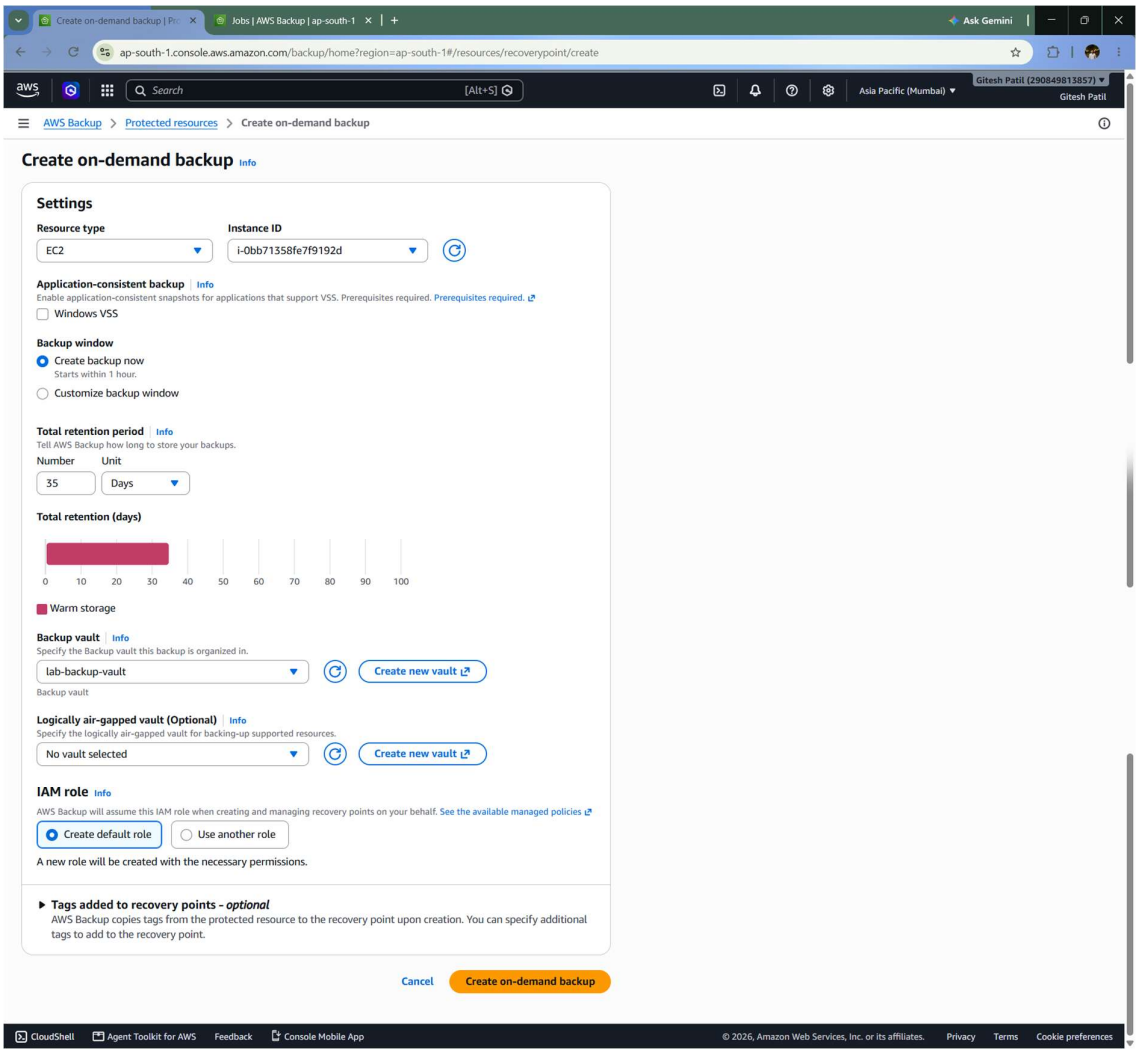
3. Click Create on-demand backup



4. Configure:

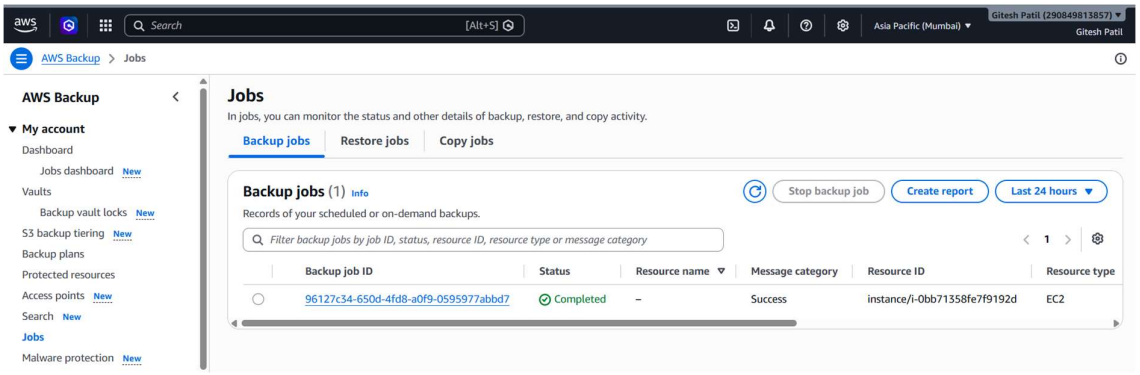
- Backup vault: lab-backup-vault
- Retention: default

5.Click Create on-demand backup



Step 10: Monitor Backup Job

1. Go to Backup jobs
2. Track job status: CREATED RUNNING COMPLETED



Step 11: Clean Up (Optional but Recommended)

1. Delete on-demand backups
2. Delete backup plan
3. Remove tags if required